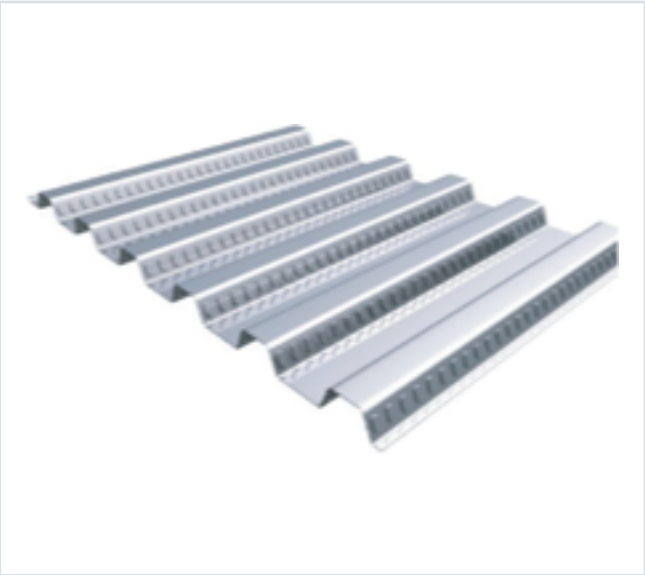


DECKING PRODUCT SUBMITTAL

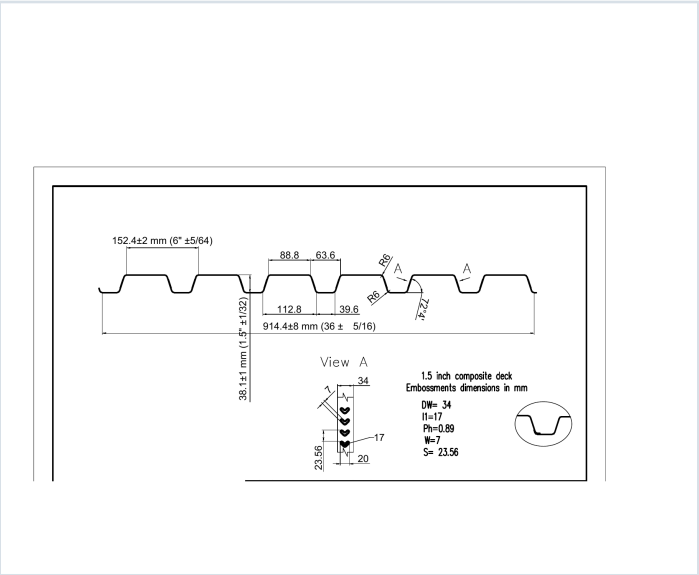
15CD-FY40-18-G90

PRODUCT	GRADE	GAUGE	COATING
1.5" Composite Deck	Grade 40	18 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 18 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0474	2.6	40	0.299	0.312	7154	7473	0.263	0.290

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 41, 42, 43, 44
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

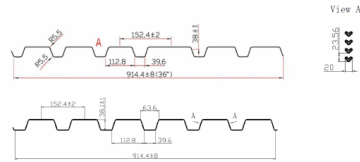
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OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 41

GRADE 40 STEEL



Section Properties

Gage	Depth (inches)	Weight lb/ft	E ksi	S _x (in ³) per foot	S _y (in ³) per foot	ASD (d = 1/8") M _t / O inch-lbs per foot	U _t (in ²) per ft.	U _y (in ²) per ft.
22	0.0096	16	40	0.173	0.094	433	443	0.147
20	0.0098	20	40	0.219	0.231	524	533	0.197
18	0.0074	24	40	0.299	0.332	756	747	0.263
16	0.0098	30	40	0.383	0.390	966	933	0.350

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/S3C RD-2017, AISI S100-2017 and AISI S100-2016.

Shear and Web Crippling

Gage	V _u / O (lb/ft)	Web Crippling (R _u) ₁ / O One Flange Loading End Bearing	Web Crippling (R _u) ₂ / O One Flange Loading Interior Bearing
22	1939	640	850
20	3042	865	1002
18	4025	1030	1192
16	4975	1245	1447

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/S3C RD-2017, AISI S100-2017 and AISI S100-2016.

Allowable Uniform Downward Loads, ASD (PSF)

Span	Gage	5'-0"	6'-0"	8'-0"	10'-0"	12'-0"	14'-0"	16'-0"	18'-0"	20'-0"	22'-0"	24'-0"	26'-0"	28'-0"	30'-0"
Single	22	110	98	77	65	56	49	43	38	34	31	28			
	20	140	116	97	83	71	62	55	48	43	39	35			
	18	158	132	107	92	80	70	62	55	49	45	41			
	16	244	202	170	145	125	109	95	85	75	68	63			
Double	22	168	147	122	102	87	75	66	58	51	46	41			
	20	248	206	173	147	127	111	97	86	77	69	62			
	18	269	226	192	163	141	123	108	96	86	77	70			
	16	405	338	288	245	212	185	162	144	128	115	104			
Triple	22	147	122	102	87	75	65	57	51	45	41	37			
	20	244	202	170	145	125	109	95	85	75	68	63			
	18	269	226	192	163	141	123	108	96	86	77	70			
	16	381	317	267	226	194	169	147	130	116	104	94			

Note: 1. All section properties and ASD (d = 1/8") uniform loads are calculated in accordance with ANSI/S3C RD-2017, AISI S100-2017 and AISI S100-2016.
2. Loads shown in table are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span conditions.
3. Bending Moment formula used for fixed-end stress limitations are: Single and Two Span: $M = \frac{wL^2}{8}$; Three Span or More: $M = \frac{wL^2}{16}$
4. Web crippling and clear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

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Span	Gage	5'-0"	6'-0"	8'-0"	10'-0"	12'-0"	14'-0"	16'-0"	18'-0"	20'-0"	22'-0"	24'-0"	26'-0"	28'-0"	30'-0"
Single	22	77	58	45	35	28	23	19	16	13	11	10			
	20	98	74	57	45	36	29	24	20	17	14	12			
	18	139	104	80	63	50	41	34	28	24	20	17			
	16	194	138	106	84	67	54	45	37	31	27	23			
Double	22	168	147	122	102	87	75	66	58	51	46	41			
	20	248	206	173	147	127	111	97	86	77	69	62			
	18	269	226	192	163	141	123	108	96	86	77	70			
	16	405	338	288	245	212	185	162	144	128	115	104			
Triple	22	147	122	102	87	75	65	57	51	45	41	37			
	20	244	202	170	145	125	109	95	85	75	68	63			
	18	269	226	192	163	141	123	108	96	86	77	70			
	16	346	280	230	194	166	143	125	110	97	86	78			

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.67.

Construction Span Table – 20 psf Construction Load

Total Slab Depth	Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
	Deck Type	Maximum Unshored Clear Span	Span	Span	Deck Type	Maximum Unshored Clear Span	Span	Span
350 (h=200) 31 PSF	15x10x22 ga	5'-4"	6'-0"	6'-0"	15x10x22 ga	5'-4"	6'-0"	6'-0"
	15x10x18 ga	6'-0"	7'-0"	7'-0"	15x10x18 ga	6'-0"	7'-0"	7'-0"
	15x10x16 ga	6'-0"	7'-0"	7'-0"	15x10x16 ga	6'-0"	7'-0"	7'-0"
	15x10x14 ga	6'-0"	7'-0"	7'-0"	15x10x14 ga	6'-0"	7'-0"	7'-0"
400 (h=250) 37 PSF	15x10x22 ga	5'-4"	6'-0"	6'-0"	15x10x22 ga	5'-4"	6'-0"	6'-0"
	15x10x20 ga	6'-0"	7'-0"	7'-0"	15x10x20 ga	6'-0"	7'-0"	7'-0"
	15x10x18 ga	6'-0"	7'-0"	7'-0"	15x10x18 ga	6'-0"	7'-0"	7'-0"
	15x10x16 ga	6'-0"	7'-0"	7'-0"	15x10x16 ga	6'-0"	7'-0"	7'-0"
450 (h=300) 43 PSF	15x10x22 ga	5'-4"	6'-0"	6'-0"	15x10x22 ga	5'-4"	6'-0"	6'-0"
	15x10x20 ga	6'-0"	7'-0"	7'-0"	15x10x20 ga	6'-0"	7'-0"	7'-0"
	15x10x18 ga	6'-0"	7'-0"	7'-0"	15x10x18 ga	6'-0"	7'-0"	7'-0"
	15x10x16 ga	6'-0"	7'-0"	7'-0"	15x10x16 ga	6'-0"	7'-0"	7'-0"
500 (h=350) 49 PSF	15x10x22 ga	5'-4"	6'-0"	6'-0"	15x10x22 ga	5'-4"	6'-0"	6'-0"
	15x10x20 ga	6'-0"	7'-0"	7'-0"	15x10x20 ga	6'-0"	7'-0"	7'-0"
	15x10x18 ga	6'-0"	7'-0"	7'-0"	15x10x18 ga	6'-0"	7'-0"	7'-0"
	15x10x16 ga	6'-0"	7'-0"	7'-0"	15x10x16 ga	6'-0"	7'-0"	7'-0"
550 (h=400) 55 PSF	15x10x22 ga	5'-4"	6'-0"	6'-0"	15x10x22 ga	5'-4"	6'-0"	6'-0"
	15x10x20 ga	6'-0"	7'-0"	7'-0"	15x10x20 ga	6'-0"	7'-0"	7'-0"
	15x10x18 ga	6'-0"	7'-0"	7'-0"	15x10x18 ga	6'-0"	7'-0"	7'-0"
	15x10x16 ga	6'-0"	7'-0"	7'-0"	15x10x16 ga	6'-0"	7'-0"	7'-0"
600 (h=450) 61 PSF	15x10x22 ga	5'-4"	6'-0"	6'-0"	15x10x22 ga	5'-4"	6'-0"	6'-0"
	15x10x20 ga	6'-0"	7'-0"	7'-0"	15x10x20 ga	6'-0"	7'-0"	7'-0"
	15x10x18 ga	6'-0"	7'-0"	7'-0"	15x10x18 ga	6'-0"	7'-0"	7'-0"
	15x10x16 ga	6'-0"	7'-0"	7'-0"	15x10x16 ga	6'-0"	7'-0"	7'-0"

Note: Web crippling and clear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

PUBLISHED PERFORMANCE DATA

15CD-FY40-18-G90 · 18 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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22 ga Normalweight Concrete (145 pcf, $f'c = 3,000$ psi)

Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	31	400	354	295	249	212	182	158
4	37	400	400	372	334	298	231	200
4.5	43	400	400	400	383	326	281	244
5	49	400	400	400	400	387	331	290
5.5	55	400	400	400	400	400	387	338
6	61	400	400	400	400	400	400	383

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	138	121	107	95	84	75	67	60
4	175	154	135	120	107	96	86	77
4.5	215	188	166	147	131	118	106	95
5	254	225	197	173	156	142	128	115
5.5	294	259	229	204	180	163	148	132
6	335	296	262	233	208	186	168	151

20/18/16 ga Normalweight Concrete (145 pcf, $f'c = 3,000$ psi)

Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	31	400	400	359	313	259	223	194
4	37	400	400	400	383	328	283	248
4.5	43	400	400	400	400	400	345	310
5	49	400	400	400	400	400	400	357
5.5	55	400	400	400	400	400	400	400
6	61	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	170	150	132	118	105	94	85	76
4	215	190	168	150	134	120	108	98
4.5	263	232	206	183	164	147	133	120
5	313	278	245	218	195	176	158	143
5.5	364	321	285	254	227	204	184	167
6	400	387	326	290	260	234	211	191

Note
In case of the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gauge. However, the construction spans do get longer for 18 and 16 gauge deck.

STEEL CATALOG PAGE 44

Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	341	285	241	206	178	154
4	28	400	400	361	308	261	225	196
4.5	33	400	400	400	372	319	275	239
5	37	400	400	400	400	379	327	285
5.5	42	400	400	400	400	400	380	332
6	46	400	400	400	400	400	400	379

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	135	119	106	94	84	75	68	61
4	172	150	136	120	107	96	87	78
4.5	210	185	164	146	131	118	106	96
5	250	221	196	175	157	141	127	115
5.5	291	257	228	204	182	164	148	134
6	333	294	261	233	209	188	170	154

20/18/16 ga Lightweight Concrete (115 pcf, $f'c = 3,000$ psi)

Slab Thickness (inches)	Weight (psf)	5'-0"	5'-6"	6'-0"	6'-6"	7'-0"	7'-6"	8'-0"
3.5	23	400	400	344	291	249	216	188
4	28	400	400	400	378	321	274	238
4.5	33	400	400	400	400	388	335	292
5	37	400	400	400	400	400	400	349
5.5	42	400	400	400	400	400	400	400
6	46	400	400	400	400	400	400	400

Slab Thickness (inches)	8'-6"	9'-0"	9'-6"	10'-0"	10'-6"	11'-0"	11'-6"	12'-0"
3.5	160	142	129	115	104	93	84	76
4	201	185	165	147	132	119	109	99
4.5	247	221	202	182	165	148	132	119
5	294	271	241	216	194	174	158	143
5.5	342	316	281	251	226	203	184	167
6	400	382	322	288	259	235	211	192

Note
In case of the profile of the embossments, there is no gain in strength for the composite deck-slab when the deck gets thicker than 20 gauge. However, the construction spans do get longer for 18 and 16 gauge deck.